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Appendix C.2. Closed–Metric Route: Rank–One Deformation Equals rank-one determinant route To $\mathcal{O}(\mathcal{T}^2)$

Take the rank–one deformation $\tilde{g}_{\mu\nu} = g_{\mu\nu} + \frac{2}{3}\lambda\,\tau\,\hat{n}_\mu\hat{n}_\nu$ so that $\sqrt{-\tilde{g}} = \sqrt{-g}\sqrt{\det[\mathbf{1} + \frac{2}{3}\lambda\,\tau\,\mathcal{T}]}$ and define $\mathcal{L}_{\rm CM} \equiv \sqrt{-\tilde{g}} - \sqrt{-g}$. The same algebra gives

$$\boxed{\delta^2\mathcal{L}_{\rm CM} = A_\star\,\sqrt{-g}\,\operatorname{sgn}(\Sigma_\epsilon)\,I_T \;+\; \nabla_\mu J^\mu_{\rm CM}, \qquad A_\star = \frac{\lambda^2}{8}.} \tag{A32}$$

Thus rank-one determinant route and CM have the *same bulk* coefficient $A_\star$ and differ only by improvements.

Appendix C.3. Pt–Even CS/Nieh–Yan Shadow: Quadratic Reduction (σ_ϵ)

Using $T^A \wedge T_A = d(e^A \wedge T_A) - e^A \wedge e^B \wedge R_{AB}$, applying $*$ and the scalar PT projector (A2), and evaluating after C1, the PT–even piece reduces at quadratic order to

$$\boxed{\delta^2\mathcal{L}_{\rm CS^+} = A_\star\,\sqrt{-g}\,\operatorname{sgn}(\Sigma_\epsilon)\,I_T \;+\; \nabla_\mu J^\mu_{\rm CS^+}, \qquad A_\star = \frac{\lambda^2}{8},} \tag{A33}$$

where the Nieh–Yan assignment (A5) reshuffles only boundary conventions inside the PT–even sector.

Appendix C.4. A Universal Canonical Improvement at Quadratic Order

For route–by–route flux comparisons it is convenient to select the *same* improvement representative for all routes:

$$\boxed{J^\mu_{\rm can}[h^{\rm TT}] \;\equiv\; \frac{A_\star}{2}\,a^3\Big(h^{\rm TT}_{ij}\,\nabla^\mu h^{\rm TT}_{ij} - \delta^\mu{}_0\,h^{\rm TT}_{ij}\,\dot{h}^{\rm TT}_{ij}\Big),} \tag{A34}$$

so that we *adopt the convention*

$$\delta^2\mathcal{L}_X = A_\star\,\sqrt{-g}\,\operatorname{sgn}(\Sigma_\epsilon)\,I_T \;+\; \nabla_\mu J^\mu_{\rm can}, \qquad X \in \{rank-onedeterminantroute, \mathrm{CM}, \mathrm{CS}^+\}. \tag{A35}$$

Different representatives differ by $\nabla_\nu X^{[\mu\nu]}$ and yield identical integrated fluxes under A4.

Check (FRW).

On spatially flat FRW in TT gauge,

$$\nabla_\mu J^\mu_{\rm can} = \frac{A_\star\,a^3}{2}\Big(\dot{h}^{\rm TT}_{ij}\dot{h}^{\rm TT}_{ij} - \frac{\partial_k h^{\rm TT}_{ij}\partial_k h^{\rm TT}_{ij}}{a^2}\Big) + \partial_\tau\big(\tfrac{3}{2}A_\star a^2\dot{a}\,h^{\rm TT}_{ij}h^{\rm TT}_{ij}\big), \tag{A36}$$

i.e., the canonical reshuffling between TT kinetic/gradient bilinears plus a pure time boundary term that integrates to zero with A4 fall–offs.

Appendix C.5. Closed Forms for FRW and Weak Field

FRW.

For $ds^2 = a(\tau)^2(-d\tau^2 + d\mathbf{x}^2)$ and homogeneous $\epsilon(\tau)$,

$$J^0_{\rm can} = \frac{A_\star}{2}\,a^3\,h^{\rm TT}_{ij}\,\dot{h}^{\rm TT}_{ij}, \qquad J^i_{\rm can} = \frac{A_\star}{2}\,a\,h^{\rm TT}_{jk}\,\partial^i h^{\rm TT}_{jk}, \tag{A37}$$

which satisfies (A36).

Weak field (AF).

At $a \to 1$,

$$J^0_{\rm can} = \frac{A_\star}{2}\,h^{\rm TT}_{ij}\,\dot{h}^{\rm TT}_{ij}, \qquad J^i_{\rm can} = \frac{A_\star}{2}\,h^{\rm TT}_{jk}\,\partial^i h^{\rm TT}_{jk}. \tag{A38}$$

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